MILITARY SPECIFICATION

BOOTS, SAFETY, AIRCREW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements for leather lined aircrew boots having water resistant leather uppers, fuel resistant soles and heels, speed lacing, and steel safety toes.

1.2 Classification The boots shall be of one type in the following whole and half sizes, and widths as specified (see 6.2):

Sizes	Widths
4 through 14 1/2	XN - Extra narrow
4 through 14 1/2	N - Narrow
4 through 14 1/2	R - Regular
4 through 14 1/2	W - Wide
4 through 14 1/2	XW - Extra Wide

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATION

FEDERAL

C-F-206	- Felt Sheet: Cloth, Felt, Wool, Pressed
V-L-61	- Laces, Nylon
V-T-285	- Thread, Polyester

Beneficial comments (recommendations, additions, and deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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V-T-295	- Thread, Nylon
KK-I-570	- Insole, Footwear, Leather, Cattlehide
TT-C-490	- Cleaning Methods and Pre-treatment of Ferrous Surfaces for Organic Coatings
TT-S-735	- Standard Test Fluids Hydrocarbon
DDD-T-86	- Tape, Textile, Cotton, General Purpose (Unbleached, bleached or dyed)
PPP-B-636	- Boxes, Shipping, Fiberboard

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MIL-S-4383	- Sealing Compound, Topcoat, Fuel Tank, Buna-N-Type
MIL-L-10867	- Leather, Cattlehide, Gusset, Chrome Tanned, Fat-liquored
MIL-P-12420	- Plastic Material, Cellular, Elastomeric
MIL-C-13924	- Coating, Oxide, Black, For Ferrous Metals
MIL-B-21408	- Boots, Safety
MIL-F-21840	- Fastener Tapes, Hook and Pile, Synthetic
MIL-S-22777	- Soles and Heels, Rubber, Traction Tread, Shoe
MIL-C-41814	- Counter, Footwear
MIL-L-43585	- Lasts, Footwear, Shoe, Safety Toe, Men's US MIL-7

STANDARDS

FEDERAL

FED-STD-151	- Metals; Test Methods
FED-STD-191	- Textile Test Methods
FED-STD-311	- Leather, Methods of Sampling and Testing
FED-STD-601	- Rubber; Sampling and Testing
FED-STD-751	- Stitches, Seams and Stitchings

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents and publications The following other Government documents and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

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2.2 Other publications The following documents form a part of this specification to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

B-36	- Brass Plate, Sheet, Strip, and Rolled Bar
B-134	- Brass, Wire
D-412	- Tension testing of Vulcanized Rubber
D-746	- Brittleness Temperature of Plastics and Elastomers by Impact
D-790	- Tests for Flexural Properties of Plastics and Electrical Insulating Materials
D-792	- Test for Specific Gravity and Density of Plastics by Displacement
D-816	- Rubber Cements
D-1084	- Viscosity of Adhesives
D-1610	- Conditioning Leather of Leather Products for Testing
D-1708	- Test for Tensile Properties of Plastics by use of Microtensile Specimens
D-2098	- Dynamic Water Resistance of Shoe Upper Leather by the Dow-Corning Leather Tester
D-2099	- Dynamic Water Resistance of Shoe Upper Leather by the Maeser Water Penetration Tester
D-2240	- Indentation of Rubber by Means of a Durometer
D-2813	- Sampling Leather for Physical and Chemical Tests

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

American National Standards Institute ANSI Z41.1-1972 - American National Standards for Men's Safety-Toe Footwear

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Association, ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, ILL 60606).

3. REQUIREMENTS

3.1 Guide sample Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

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3.2 First article When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval.

3.3 Materials

3.3.1 Upper leather The uppers shall be cut from the best quality green salted, chrome tanned, full grain sides, with the thickness not less than 4 1/2 ounces nor more than 6 ounces, however, the backstays may be a minimum of 3 1/2 ounces. The area of a side shall not exceed 28 square feet. The tannery lot shall not exceed 25,000 square feet. The sides shall be drum dyed black and finish grain side out with no buffing or snuffing to remove surface blemishes. The contractor shall supply a certificate of compliance stating that a top finish was not applied to achieve the water resistance requirements of 3.3.1.1.1 and 3.3.1.1.2. Embossing or printing of the leather shall not be permitted (see 4.3.1). The break in the leather shall not be more than a Break Scale of 5 (see 4.3.5.1). The cut parts shall be examined as specified in 4.3.2.1. Upper leather defects such as fat wrinkles, pits, stretchy leather, or insect damage shall not appear in the vamps unless barely perceptible. Cut parts with scratches, slaughter cuts that show through on the grain surface of the leather shall not be used. Flanky, fleshy, loose, boney, pipey, cracked, or rough or coarse grain leather shall not appear in the vamps, quarters, backstays or tongues.

3.3.1.1 Water resistance The upper leather shall be treated with a suitable water resistance compound to resist penetration and absorption by water. The leather shall be tested as follows:

3.3.1.1.1 Water penetration The leather shall withstand 4000 Dow Corning flexes or 15,000 Maeser flexes; however, 30 percent of the samples shall withstand not less than 1000 Dow Corning flexes or 3000 Maeser flexes when tested as specified in 4.3.1.

3.3.1.1.2 Water absorption The leather shall absorb not more than 30 percent of its weight after being immersed in water when tested as specified in 4.3.1.

3.3.1.2 Fungi resistance Leather components do not require a fungicide treatment. However, when used, leather components shall contain not more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.3.1.

3.3.2 Tongues, speed lacer stays, and top band cuffs The leather for the tongues, speed lacer stays, and top band cuffs shall conform to treatment B of MIL-L-10867, except that all the specimens tested for water resistance shall withstand a minimum of 50 taps and the requirements for stitch tearing strength, shrinkage, paranitrophenol, stiffness, and chemical requirements shall not apply. The water resistant requirement is not applicable to the top band cuffs or speed lacer stays. Leather specified in 3.3.1 may be used for the speed lacer stays and top band cuffs, except that the thickness shall be 2 3/4 to 3 1/2 ounces.

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3.3.3 Lining, leather The lining leather shall be soft, chrome tanned, full grain or corrected grain, cattlehide, glove or pigskin leather of 2 to 3 1/2 ounce thickness. The color of the leather shall be as produced by the tanning agents. A light application of dye may be used to level the color of the grain surface. Flesh cuts and flanky or pipey leather which may affect serviceability will not be permitted.

3.3.4 Heel pads The leather for the heel pads shall be as specified in 3.3.3 or shall be sheepskin, kid, calf, or kip with a thickness of 2 to 3 1/2 ounces. Heel pads cut from upper leather or tongue leather may be used provided the top finish has been removed by buffing or splitting, and the thickness is 2 to 3 1/2 ounces.

3.3.5 Insoles

3.3.5.1 Leather When leather insoles are used they shall conform to type I, class 1, tannage (a) or (b) of KK-I-570, except that the thickness shall be 5 1/2 to 7 1/2 irons and the requirements for paranitrophenol shall not apply.

3.3.5.2 Synthetic When synthetic insoles are used they shall contain 0.25 to 1.00 percent copper-8-quinolinolate uniformly distributed throughout the insole. The insole shall be 0.116 (± 0.010) inch thick and shall be cut with the heel-to-toe direction across the machine direction of the synthetic material. The synthetic insole shall have a stuck-on rib applied to the printed side of the material. The synthetic insoles shall conform to the above when tested as specified in 4.3.1 (see 6.5).

3.3.6 Welting, vinyl The vinyl welting shall be of flat top design, bright black in color, and may be formed with or without a storm welt of medium size bead. The welting shall be of sufficient width to assure that the required extensions on the finished shoes are met (see 3.7.22). The thickness of the welting shall be 0.11 to 0.13 inch. The base compound shall be virgin polyvinyl chloride compounded with suitable non-bleeding plasticizers. The finished welting shall be smooth, uniform in texture, and free of blemishes, cracks, blisters, and other defects that might affect serviceability or appearance. The vinyl welting shall conform to the requirements of table I when tested as specified in 4.3.1.

Table I - Vinyl welting requirements

Property	Requirements
Initial	
Hardness	85-95
Specific gravity	1.32 maximum
Brittle point, before aging	-15°F or less
Brittle point, after aging	+20°F maximum change from standard brittle point
Tensile strength, psi	2100 minimum
Elongation, percent	225 minimum
Water absorption, percent gain	3.00 maximum
Solvent Resistance (After Immersion)	
Hardness	55 minimum
Tensile strength, psi	1800 minimum
Elongation, percent	225 minimum
Volume swell, percent	35 maximum

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3.3.7 Counters Counters shall conform to the requirements of MIL-C-41814. As an alternate, an extruded sheet of virgin ionomer resin with heat activated adhesive on both sides may be used. The ionomer resin shall conform to the following requirements when tested as specified in Table VI (see 6.5):

Property	Requirements
Thickness (sheet), inch	0.057 to 0.063
Thickness (adhesive coating, each side) inch	0.003 to 0.006
Specific gravity, g/cc	0.940 to 0.970
Tensile strength, psi	3500 to 4500
Elongation, percent	250 (min)
Flexural modulus, psi	45,000 to 55,000
Brittleness temperature	-60°C (max)

3.3.7.1 Counter sizes Counters shall be made in a sufficient number of sizes to cover boots sized 4 thru 14 1/2 and necessary widths. Counters shall conform to the shape of the MIL-7-last. Counter patterns shall assure that counters are the correct size and shape.

3.3.8 Rubber

3.3.8.1 Outsoles The rubber outsoles shall be 16 irons in thickness and conform to the physical requirements for type I, Grade A of MIL-S-22777. As an alternate, the tread area may be of a straight bar design or of a commercial design approved by the preparing activity of this specification.

3.3.8.2 Heels The rubber heels shall conform to Type II, Grade A, Class 1, of MIL-S-22777. As an alternate, the tread area may be of a straight bar design or of a commercial design that has been approved by the preparing activity of this specification.

3.3.9 Fabrics

3.3.9.1 Doublers The doubler fabric for the vamp shall be cut from double or single napped cotton flannel, 5.65 ± 0.50 oz/sq yd. The finished fabric shall have a minimum thread count of 36 by 36, with a minimum breaking strength of 30 pounds in the warp and 30 pounds in the filling. The non-fibrous materials shall not exceed 15 percent. The color of the fabric shall be natural or bleached. Testing shall be as specified in 4.3.1.

3.3.9.2 Toe cushion The material used to cushion the steel box toe and mask the breastline, shall be of wool felt conforming to type III, classification 9A-2 of C-F-206. As an alternate material (a) a foamed polyvinyl material 3/32 (± 1/32) inch in thickness weighing 4 to 6 pounds per cubic foot or (b) a latex foam rubber 1/8 (± 1/64) inch thick may be used. The toe cushion shall be cut large enough to extend approximately 1/2 inch rearward of the steel toe breastline.

3.3.9.3 Understructure box toe The understructure box toe material shall conform to either 3.3.9.3.1 or 3.3.9.3.2.

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3.3.9.3.1 Expanded plastic sheet The understructure box toe shall be made of material conforming to Type II, Class 4 of MIL-P-12420, except for the changes below.

Density - lbs/cubic foot	5.0 to 6.5
Thickness (inch)	0.125 (± 0.020)
Tensile strength, psi (min)	
Initial	50
After aging	50
Compression deflection (psi)	2.0 min.
	3.5 max.
Compression set, percent	20 max.

3.3.9.3.2 Laminated sheet When automated lasting is used the understructure box toe may be made of virgin ionomer resin laminated with a combination of flannel or nonwoven fabric on one side and a flannel, nonwoven or sheeting fabric on the other side. The combined thickness of the lamination shall be 0.035 (±0.005) inch. Testing shall be as specified in 4.3.1. (see 6.5).

3.3.9.4 Backseam tape. The backseam tape shall be 1/2 or 17/32 inch wide conforming to type I, class 1 of DDD-T-86 except that the requirement for non-fibrous materials shall not apply.

3.3.9.5 Fastener tape The fastener tape for securing the top cuff to the top of the boot shall be 5/8 to 3/4 inch wide and 2 to 2 1/8 inches long. The color shall be black. The hook and pile fastener tape shall conform to type I, class 1 of MIL-F-21840.

3.3.10 Metals

3.3.10.1 Steel box toe The steel box toe shall conform to the toe of the last and shall be fabricated from cold-rolled carbon steel, and shall conform to the requirements of Table II after heat treating when tested as specified in 4.3.1. The steel box toes shall be thoroughly cleaned and completely coated with a zinc compound. As an alternate, the steel toes may be coated with any suitable coating that will assure protection against corrosion and will not damage component parts of the boot.

Table II - Physical requirements, steel box toe

Hardness (Rockwell C Scale)	Carbon Content	Thickness (inch)	Trade pattern no.
43 to 50	0.50 to 0.82	.062 ± .0025	400

3.3.10.1.1 Impact resistance The steel box toe of the finished boot shall have a minimum inside clearance of 1/2 inch when tested as specified in 4.3.5.2.

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3.3.10.2 Eyelets The eyelets shall be of aluminum, 0.016 (± 0.0015) inch thick with a roll barrel setting. The eyelets shall be anodized or finished with a corrosive preventive treatment and the heads shall be roller or tumble coated with not less than two coats of black enamel. The eyelet barrels shall be of sufficient length to secure all parts through which they are driven. The eyelets shall have an outside diameter of 0.385 to 0.390 inches and a minimum inside diameter of 0.180 inch when tested as specified in 4.3.1.

3.3.10.3 Speed lacers The speed lacer assembly shall be a "D" shaped ring attached to a base with integral rivet. Length of the rivet shall be sufficient to make a secure clinch thru the quarter gusset and speed lacer row stay. The speed lace assembly shall be steel with a durable black enamel finish (see 6.5).

3.3.10.4 Clinch clamp The clinch clamp shall be aluminum, black in color and shall conform to figure 3 (see 6.5).

3.3.10.5 Nails, heel attaching Heel nails shall be brass, cut or wire type and of sufficient length to produce a secure clinch on the insole. The cut type nails shall be of material conforming to alloy No. 6 or No. 8 of ASTM Standard B-36 and shall be commercial No. 450 or 1336 with a 20 gauge point when gauged as specified in 4.4.3. The wire nails shall be commercial type 13 gauge with a flat clinching point and shall be of material conforming to alloy No. 7 of ASTM Standard B-134 when tested as specified in 4.3.1.

3.3.10.6 Tacks and staples. Assembly tacks and staples, heel seat lasting tacks, welt butt tacks, and tacks or staples used for attaching the shank shall be brass or steel and shall be of sufficient length to thoroughly attach the parts through which they are driven, and leave the insole smooth on the inside. All brass tacks shall be of material conforming to alloy No. 6 or No. 8 of ASTM Standard B-36 when tested as specified in 4.3.1.

3.3.10.7 Nails, heel seat fastening Heel seat fastening shall be done using No. 12, thru 39 head, brass nails or steel nails of sufficient length to secure parts through which they are driven and leave a secure, smooth clinch on the insole. Testing shall be as specified in 4.3.1.

3.3.10.8 Steel shank combination The shank combination shall be made of a shank board piece with a steel shank securely attached by four prongs or two rivets. The steel shank shall be located 1/2 to (± 1/16) inch from the front edge of the shank board. There shall be 7 sizes of shank combinations for the run of boot sizes specified in Table III.

3.3.10.8.1 Steel shank The steel shank shall be constructed from 19 gage 0.0418 (± 0.0030) inch cold rolled carbon steel with a hardness ranging from 47 to 54 Rockwell C scale when tested as specified in 4.3.1 and shaped to the manufacturer's standard number 4 bend, conforming to the arch of the MIL-7 last. The width of the shank shall be 5/8 (± 1/64) inch. The shank shall be made with a single or double rib. The overall thickness of the steel shank with rib shall be 0.080 to 0.125 inch. The steel shank shall have a zinc electroplated coating, dull or bright finish, or a zinc compound coating completely and uniformly applied to the base metal following a cleansing by any method of TT-C-490. Steel shanks

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with black oxide coating conforming to Class 1 of MIL-C-13924 may be used in lieu of shanks with zinc coating. The rib or ribs shall taper off at a point within 3/8 to 7/8 inch from each end. The steel shank shall grade up or down 1/4 (± 1/16) inch, the size 4 being 4 1/2 (± 1/16) inches long.

3.3.10.8.2 Shank board The shank board shall be water resistant (shank board that has been hot waxed is considered being water resistant). The finished thickness of the shank board shall be 1/8 (± 1/32) inch. The shank board shall be cut in accordance with Government furnished patterns specified in 3.5. The shank board pieces shall be molded to follow the bottom cavity of the boots, and both ends shall be skived with a graduated scarf 1 3/8 (± 1/16) inches wide. The shank board pieces shall be graded up or down 1/4 (± 1/16) inch and shall be marked consecutively with a steel stamp starting with figure "1" on the smallest size up thru "7" on the larger size. The size 4 shank board piece shall be 6 (± 1/16) inches long.

3.3.10.9 Fiberglass shank As an alternate, a fiberglass shank (see 6.6) without shank board may be used. The shank shall consist of glass fibers impregnated with vinyl ester resins encased in a plastic sleeve or tube. The width of the uncured encased fiberglass resins shall be 5/8 (± 1/32) inch. The shank shall be cured and bonded to the insole and conform to the shape of the last. When fully cured the shank (glass and resin area) shall be 5/8 (± 1/16) inch in width. The cured length of the shanks shall be graded as specified in 3.3.10.8.2. The forward end of the shank shall have a 1/2 (± 1/8) inch taper. The heel end shall be tapered for a minimum of 3/8 inch.

Table III Shank fitting schedule

Shank sizes	1						2				3			
Shoe widths and sizes 1/														
XN-N	4	4½	5	5½	6	6½	7	7½	8	8½	9	9½	10	10½
R - W	4	4½	5				5½	6	6½	7	7½	8	8½	9
XW	4	4½					5	5½	6	6½	7	7½	8	8½

	4				5				6			7
XN-N	11	11½	12	12½	13	13½	14		14½			-
R-W	9½	10	10½	11	11½	12	12½	13	13½	14	14½	-
XW	9	9½	10	10½	11	11½	12	12½	13	13½	14	14½

1/ Except that the XN width is not required for sizes 4 and 4½.

3.3.11 Thread

3.3.11.1 Thread, upper fitting Upper fitting stitching operations shall be performed using nylon thread conforming to Type I, II or III of V-T-295. Colorfastness requirements shall not apply. Thread size shall be E for the needle and bobbin on stitch type 301 and E for the needle and B or E for the looper with stitch type 401. The color shall be black.

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3.3.11.2 Thread, Goodyear stitching Thread used for Goodyear stitching shall be polyester conforming to Type I, Class 1, sub-class C of V-T-285. The running thread shall be black, size 10, 3 ply. The shuttle thread shall be natural, size 10, 3 or 6 ply. Colorfastness requirements shall not apply.

3.3.11.3 Thread, inseaming The inseaming thread shall be polyester conforming to type I, class 1, subclass C of V-T-285. The thread shall be size 10, 3 or 6 ply, natural color.

3.3.12 Bottom filler The bottom filler shall be either thermoplastic or cold process type conforming to the following requirements:

3.3.12.1 Thermoplastic bottom filler The thermoplastic type shall consist of a mixture of ground cork and a suitable thermoplastic binder in the proportion of a minimum of 2 3/4 parts by volume of cork to each part of binder. The cork granules shall be of the best quality and free of bark. The ground cork and binder shall be thoroughly and evenly mixed. The binder shall be water insoluble and flexible. It shall have a softening point of at least 125°F and a maximum penetration of 85 millimeters at a 200 gram load for 50 seconds at 77°F when tested as specified in 4.3.1.

3.3.12.2 Cold process bottom filler The cold process type shall be spreadable without the use of heat. It shall consist of a mixture of ground cork and a suitable binder in the proportion of two parts cork to each part of binder by volume. The cork granules shall be free from bark. When the spread filler is dry and set, it shall consist of four parts cork to one part binder. The binder shall be water insoluble and have a softening point of at least 150°F (66°C). Testing shall be as specified in 4.3.1. As an alternate, a cold process bottom filler applied by semi-automatic equipment or other suitable method may be used. The alternate filler shall consist by weight of a mixture of one part ground cork to five parts suitable binder. Upon loss of solvent, the alternate filler shall consist by weight of one part cork to 3 3/4 parts binder. The binder shall be water-insoluble and have a softening point of at least 125°F (51.8°C) when tested as specified in 4.3.1.

3.3.13 Laces, nylon Laces shall conform to type II, class 1 of V-L-61. The color shall be black and the laces shall have a minimum length of 72 inches.

3.3.14 Adhesives

3.3.14.1 Seam seal adhesive The adhesives used for sealing the upper seams and inseam areas shall be as specified in 3.3.14.1.1 or 3.3.14.1.2.

3.3.14.1.1 Vinyl type The adhesive shall be composed of a polyvinyl butryal base, a non-bleeding pthalate plasticizer, and other ingredients necessary to produce a clear colorless seam sealant. The adhesive shall have a minimum weight of 7.40 pounds per gallon, a minimum total solids content of 19 percent and a viscosity of 40 to 50 seconds when tested as specified in 4.3.1.

3.3.14.1.2 Buna-N-type The adhesives shall conform to the requirements of MIL-S-4383 except that the color shall be natural or colorless.

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3.3.14.2 Adhesive, general The adhesives used for bonding various other parts of the boot may be any suitable water resistant rubber or synthetic resin adhesive. Testing shall be as specified in 4.3.1.

3.3.15 Wax The wax used during the inseam sewing and Goodyear stitching operations shall be white or golden in color and shall be a permanently plasticized resin that will thoroughly penetrate the thread at temperatures encountered during normal sewing machine use.

3.3.16 Cleaner solution A suitable cleaning solution shall be used to remove grease and soil marks of manufacturing operations from the boot uppers. The solution shall have a maximum pH value of 80 and shall not decrease the water resistant characteristics of the upper leather when tested as specified in 4.3.1.

3.3.17 Repairers Repairers shall be a liquid spray, crayon or paste type applied by hand. The color shall match the color of the upper leather and shall have sufficient coverage to correct minor surface imperfections of the leather.

3.3.18 Renovators Renovators used in lieu of, or in addition to repairers shall match the color of the upper leather. Application may be by sponge or spray method.

3.3.19 Insole, stuck-on rib The finished rib shall consist of a combination of coated fabric and fiberboard materials. The rib shall be 7/32 to 8/32 inch high with a minimum width of 5/8 inch when measured from the inside vertical portion of the rib, and shall extend around the periphery of the insole from score line to score line, which is 2 5/8 inches from the end of heel on a size 8R insole; other sizes shall be graded up and down to insure proper welt butt location on the finished last. The fabric used for the stuck-on rib shall be cotton or cotton synthetic conforming to the requirements listed below. The fabric shall be coated on one side with a suitable adhesive and bonded to the flesh side of the insole. The fiberboard and fabric shall cover 4/32 to 6/32 inch of the peripheral edge, and provide for the required edge extension of the finished boot.

Fabric requirements

Weight, ounces per sq. yd (min)	Yarns per inch (min)		Breaking strength, lbs. (min) warp and filling
	Warp	Filling	
8.0	52	30	105

3.3.19.1 Rib strength The physical requirements for the stuck-on rib shall conform to the list below when tested as specified in 4.3.4.2.

Rib strength requirements

Characteristic	Minimum 1/	Average 1/
Shear strength, lbs	70	75
Stitch strength, lbs.	20	30

1/ No single determination shall fall below the minimum specified and the average of all specimens shall not be less than the average specified.

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3.4 Design The design shall be for a blucher style, nine inch high boot with a fold down top cuff, speed lace closure with clinch clamp, safety steel box toe, one piece vamp, combination tongue and gusset, backstay, and leather lined vamp and quarters. The boot shall have a full length synthetic rubber sole with a whole heel and synthetic welting. (see figures 1 thru 4).

3.5 Patterns and dies A standard set of paper patterns and markers will be furnished by the Government (see 6.2) as a basis from which the contractor's dies or patterns shall be made. The patterns, markers and contractor's cutting dies and patterns shall consist of the component parts specified in Table IV. The Government patterns may be slightly modified for stitching and lasting ease when approved by the procuring activity. The lasting allowance shall be determined by the contractor.

Table IV - Patterns and markers 1/

COMPONENT PARTS	
Vamp	Whole and half sizes, all widths
Quarters (right and left)	Whole and half sizes, all widths
Back stay	Whole and half sizes, all widths
Tongue and gusset	Whole sizes, all widths
Quarter linings (right and left)	Whole sizes, all widths
Vamp lining	Whole sizes, all widths
Vamp doubler	Whole sizes, all widths
Eyelet/speed lace stay	Whole sizes, all widths
Top facing cuff	Whole sizes, two widths 2/
Shankboard	Seven sizes
MARKERS	
Vamp	Whole and half sizes, all widths
Quarters	Whole and half sizes, all widths

1/ The boots shall be made with the patterns specified for MIL-B-21408, Boot, Safety, with the exception that a top facing cuff will replace the pattern for the top band.

2/ The Government shall furnish a pattern for the top facing cuff in size 9R and the contractor shall be responsible for grading, up and down, all other sizes and widths.

3.6 Lasts and insole patterns The boots shall be made on Government lasts (MIL-L-43585, MIL-7). The Government will loan the contractor lasts and paper insole patterns for each size and width specified (see 6.2). The contractor shall furnish, if utilized, the wood insole rounding patterns or dies, conforming to the paper insole patterns loaned by the Government.

3.7 Construction

3.7.1 Cutting of uppers The uppers shall be cut from grain out leather specified in 3.3.1. Vamps shall be cut from the bend area of the side leather. Except for speed lace stay, top facing cuff and back stay none of the parts shall be split or shaved. Leather parts cut off stretch shall not be used.

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3.7.2 Skiving Skived edges shall be sufficient to provide a smooth, even fit of the component parts. All skiving shall be done on the flesh side in the following locations:

Part	Location of skiving
Uppers	Throat and wings
Vamp lining	Throat and wings
Quarter linings	Vamp lap and throat

3.7.3 Doubler attachment The cotton flannel doublers used for leather reinforcement shall be smoothly and securely cemented to the flesh side of the leather vamp with a suitable adhesive that will not harden the leather.

3.7.4 Upper leather fitting and seam sealing

3.7.4.1 Seam sealing Before upper fitting, paint all edges that are to be stitched, except the quarter and quarter lining backseams, with a seam sealer as specified in 3.3.14.1. While tacky, join, press and stitch the uppers as specified in 3.7.4.2. As an alternate, the quarters, vamp, and backstay may be stitched together and their finished seam stitching and overlap painted on the inside with seam sealer.

3.7.4.2 Upper leather fitting types Line marker patterns shall be used for all upper fitting. The leather shall not be die stamp marked. All stitching shall conform to FED-STD-751. The vamp and quarter linings shall be assembled and sewn together with two rows of stitching, 1/16 to 3/32 inch apart, using stitch type 301 with 8 to 10 stitches per inch. The first row shall be 1/16 inch from the edge of the leather. The quarter linings shall be closed at the back with stitch type 401 with 8 to 10 stitches per inch. The lining back seam shall be rubbed flat and reinforced on the flesh side with the tape specified in 3.3.9.4. The tape shall be centered over the seam and attached with one row of stitching on each side, using stitch type 301 with 8 to 10 stitches per inch. The vamp and quarters shall be closed at the back by butting the edges and using stitch type 502 with 8 to 10 stitches per inch. The back seam area shall be flat and smooth and shall not contain any wrinkles or bunching. The quarters shall be attached to the vamp with three or four rows of stitching with a single or double needle 1/16 (± 1/32) inch apart, using stitch type 301 with 10 to 12 stitches per inch. The first row shall be 1/16 inch from the edge of the leather. The back stay shall be centered over the backseam and attached with two rows of stitching with a single or double needle on each side, 1/16 (± 1/32) inch apart, using stitch type 301 with 10 to 12 stitches per inch. The first row shall be 1/16 inch from the edge of the leather. The back stay shall be centered over the backseam and attached with two rows of stitching with a single or double needle on each side, 1/16 (± 1/32) inch apart, using stitch type 301 with 10 to 12 stitches per inch. The first row shall be 1/16 inch from the edge of the leather. The band portion of the combination top band and cuff piece shall be sewn to the inside top edge of boot with two rows of 5/8 inch gauge stitching using stitch type 301 and 8 to 10 stitches per inch. The top band cuff shall be jointed to the boot after the quarter lining and speed lacer stay piece have been sewn to the quarter. The outer row of the speed lacer stay stitching shall

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be accomplished on a machine equipped with a knife for undertrimming the gusset and speed lacer stay leather. The inner row of speed lacer stay stitching shall be 3/4 (+ 1/16) inch from the outer row. Vamps with attached doublers shall be crimped not more than one pair at a time. The vamp lining and combined vamp/doubler shall be sewn to the vamp at the throat using stitch type 301 with 8 to 10 stitches per inch. The one piece combination tongue and gussets shall be creased and pounded along the fold where it forms the gusset. Stitch the combination tongue/gusset piece to the vamp and vamp lining at the throat with two rows of crescent shaped stitching. The first row of stitching shall follow the curvature of the tongue/gusset at the throat and the second row shall follow the curvature of the vamp lining at the throat. To prevent leakage the smallest size needle shall be used.

NOTE: In lieu of vamp lining and quarter lining assembly and assembly of linings to uppers specified above, the vamp lining, tongue and vamp may be sewn together in one operation according to vamp marker, with two rows of stitching from the outside at the vamp throat with the stitching close to the edge of the leather. One row of stitching from the inside shall be used to stitch the vamp lining to the tongue and shall be close to the edge of the leather. The quarter lining shall then be inserted in the uppers and cemented together at the vamp lining seam and stitched together with two rows of stitching 1/16 to 3/32 inch apart. The first row shall be approximately 1/16 inch from the edge of the leather.

3.7.4.3 Coating and sealing, vamp, tongue and barring connection Coat the inside upper stitching or bobbin threads across the vamp-throat-tongue and at barring connections with a light coat of seam sealer (see 3.3.14.1). Seal all outside needle threads with concentrated alkenyl succunic acid (Bavon) or 15 percent Sylmer solution (a silicone polymer). Seal stitching of speed lacer facings.

3.7.5 Eyeletting and speed lacers Seven eyelets shall be fastened to the boot in accordance with contractor supplied marker patterns or any other suitable positioning method. Speed lacers shall be inserted thru the top six eyelets and clinched on the underside of the six eyelets. Each quarter shall contain six speed lacers and one eyelet. The outer edge of the bottom eyelet shall be not less than 3/16 inch or more then 5/16 inch from edge of quarter. The outer front edge of lacer ring shall finish even with or project not more than 1/8 inch beyond edge of quarter. The top lacer shall be positioned just far enough below the bottom edge of top facing to allow the rivet to miss going thru the facing piece. The bottom eyelet shall be positioned so that its outer edge is approximately 1/4 inch from edge of quarter at the blucher. The remaining speed lacers shall be evenly spaced and properly aligned.

3.7.6 Understructure box toe The understructure box toe shall be cut from material specified in 3.3.9.3. The understructure box toe shall be cut large enough to completely line the steel box toe, extend approximately 1/2 inch rearward of the box toe breast line and be large enough to be caught in the inseaming operation.

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3.7.7 Lacing for lasting The quarters shall be machine laced thru three or four lacing holes. After lasting, the distance between the quarters shall be 3/4 to 1 inch.

3.7.8 Barring The blucher ears shall be barred with a rectangular box stitch using stitch type 301 with 10 to 12 stitches per inch. The box stitch shall be 5/8 ± 1/16 inch in length and parallel to the side seam stitching. The width of the box stitch shall be 1/8 to 3/8 inch and positioned between the top row of side seam stitching and inner row of lacer stay stitching (see figure 1).

3.7.9 Leather insoles When used, leather insoles shall be lightly degrained or heavily buffed and shall be properly fleshed. The insoles shall be rounded to pattern. Die cut insoles will be accepted provided they conform to paper insole patterns loaned by the Government. A ± 1/64 inch tolerance from Government loaned insole patterns will be allowed.

3.7.10 Lasting Uppers may be conditioned by any suitable means except that they shall not be dipped in water. The correct size and width of uppers, box toe and insoles shall be assembled to the last. Insoles shall be attached to the last with not less than 5 tacks or staples; one in the center of the heel seat, one at the shank, one at each side of the ball, and one at the toe. Edges of the insole shall be even with the last bottom at all points. Counters shall be coated on both sides with any suitable water resistant adhesive. Counters shall be sized to fill the counter pocket and large enough to provide for a wiped in heel seat of not less than 1/2 inch or more than 9/16 inch. As an alternate, heel seat lasting may be done by using any suitable method or equipment that assures a flat, secure heel seat. The foam understructure box toe specified in 3.3.9.3 and 3.7.6, shall be cemented to the flesh side of the vamp lining. Toe cushion material specified in 3.3.9.2 shall be positioned between the steel box toe and vamp doubler. The correct size steel box toe shall be carefully positioned between the understructure box toe and toe cushion material. The upper shall then be pulled down to the last with proper tension and blucher noses shall be even. Sides of the boot shall be spindled and uppers firmly secured to the last. As an alternate, the side lasting and inseaming operations may be performed simultaneously with any suitable equipment or method. The heel seat shall be wiped in, tacked, and be free of wrinkles. The toe shall be smoothly wiped in and secured at the base of the insole rib by wire or monofilament nylon. No tacks or staples shall be used above lasting line or in the back of the quarter. As an alternate, any suitable equipment or method may be used to smoothly and securely attach the toe around the base of the insole rib. When the alternate method is used the bottom edges of the toe cushion and understructured box toe material shall be trimmed so as not to interfere with the cementing of the lasting operation. When the alternate method is used to accommodate automatic equipment, the grain surface of the leather lining may be lightly buffed to provide an adequate surface to which the adhesive will adhere. The buffed area shall be no more than 2 inches in from the front (center of toe) edge of the lining and graduate out to the side of the toe in the toe lasted area.

3.7.11 Time allowance on lasts The boots shall remain on the lasts until all parts are thoroughly dry.

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3.7.12 Inseaming and first seam sealing A seam sealed as specified in 3.3.14.1 shall be applied over the lasted margin. Inseam the synthetic welting with thread specified in 3.3.11.3. The thread shall be thoroughly hot waxed, with proper tension on the thread, using a needle not larger than No. 41 and with not less than 3 1/4 stitches per inch. The seam shall be tight and the inseam shall be stitched from welt butt to welt butt extending to a point which will catch in the counter with one or two stitches on each side.

3.7.13 Tack or staple pulling, inseam trimming Care shall be taken that all tacks or staples have been removed and that no broken tack or staple points remain. The inseam shall be carefully and closely and evenly trimmed from the butt of welt to butt of welt without cutting or damaging the stitches. The ends of the welt shall be skived with a 5/8 (± 1/8) inch bevel and tacked within the butt area. The welt shall be joined by the inseamer with adequate tension and machine adjustment, whereas it will hinge easily after joining.

3.7.14 Second seam sealing A heavy coat of seam sealant as specified in 3.3.14.1 shall be applied over the welting, trimmed inseam, and insole areas, except heel seat area, and allowed to cure at room temperature. The covered areas shall include the feather line to top of rib, inseam stitching, nail or tack holes, tacks, and staples. No seam sealer shall be applied to the welt-vamp junction on the outside of the boot after the inseaming operation.

3.7.15 Shank fitting and bottom filling

3.7.15.1 Steel shank The properly sized shank shall be selected in accordance with the casing schedule specified in 3.3.10.8.2. The shank shall be positioned so that the shankboard cover fills the cavity between the inner edges of the insole rib from the ball line rearward to the back of the heel seat. The forward end of the shankboard cover shall be flush with the insole and shall fit the contour of the boot and secured with pitch, wax, two tacks or staples. The tacks or staples shall be inserted in the rear of the shankboard cover, one on each side of the steel shank. Any space between the edge of the shankboard cover and the insole rib shall not be more than 1/8 inch. The bottom filler shall be applied and firmly pressed into the insole around the toe and extending to the forward end of the shankboard. The bottom shall have a firm, uniform, and even surface for the sole laying.

3.7.15.2 Fiberglass shank When the alternate fiberglass shank specified in 3.3.10.9 is used, the forward end of the shank shall be positioned approximately 1/2 inch rearward of the ball line with the back end extending into the heel seat area. The shank shall be cured and bonded to the insole and shall conform to the shape of the last. The bottom filler shall be applied from the forward end of the shank back through to the heel seat, filling all cavities around the shank.

3.7.16 Sole laying The boot bottoms, except heel seat area, shall be coated with a suitable adhesive. The outsole shall be positioned on the boot and laid on a sole laying machine with pressure. The outsoles shall be of adequate size and laid evenly to allow for the specified edge extension.

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3.7.17 Rough rounding Soles shall be evenly rounded on a rough rounding machine to provide for the edge extension specified in 3.7.22.

3.7.18 Goodyear stitching The soles and welt shall be sewn together on a lock-stitch machine with the thread specified in 3.3.11.2. The threads shall be thoroughly impregnated with the wax specified in 3.3.15. The size of the needle and the awl shall not exceed 45 and the stitches per inch shall be 4 1/2 to 7, except that there shall be no more than 3 1/2 stitches per any one-half inch length in the ball and toe sole stitching. The stitching shall be laid on the surface and close to the feather edge of the welt with the starting and finishing ends of the stitchline extending under the heel on the finished boot. The Goodyear stitching shall not fall in the traction design area of the outsole. The stitching shall be aloft with the lock (peg) just under the surface of the sole.

3.7.19 Heel seat fastening Heel seat fastening shall be done using nails specified in 3.3.10.5, driven three to the inch and properly positioned at the edge of the insole from welt butt to welt butt to provide a secure, smooth clinch on the insole.

3.7.20 Heel seat rounding The outsole in the heel seat area shall be smoothly rounded from butt of welt to butt of welt.

3.7.21 Heel attaching The heels shall be attached with 13 nails specified in 3.3.10.7. The nails shall be of sufficient length to insure a smooth, secure clinch on the insole. The heeling machine shall be equipped with proper length drivers to assure that all nails are driven evenly against the plastic or composition cores in the heels and clinch the nails on the insole. Nails shall not be driven in the heel tread pattern. Properly sized heels shall be used for the full range of boots sizes.

3.7.22 Edge trimming The edges shall be trimmed square. The finished extension shall be as specified in Table V. The edge of welt, outsole and heel shall be smoothly joined.

Table V - Finished extension

Location	Requirement (inches)
Outside ball	1/4 (± 1/32)
Toe	3/16 (± 1/32)
Inside ball	3/16 (± 1/32)
Shank	3/16 (± 1/32)

3.7.23 Heel trimming and scouring The heels shall be trimmed square and smoothly scoured. The beveled breastline shall not be scoured.

3.7.24 Edge finishing The finished edges of the welt and outsole shall be clean and smooth. The use of edge stain or blackening is prohibited.

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3.7.25 Finishing

3.7.25.1 Preparation The boot uppers shall be properly cleaned and all excess wax and other foreign matter removed. The surface of the leather shall be conditioned to receive further application of a suitable silicone water resistant finish. Soles and heels shall be cleaned of excessive adhesive, thread ends, and other foreign matter.

3.7.25.2 Nails, tacks and staples Nails, tacks and staples that have been left protruding through the insole and cannot be pulled out, shall be cut close to the surface leaving no protruding stumps. A mechanical tack detector or other suitable method may be used to indicate the presence of any protruding nails, tacks or staples inside the boot.

3.7.25.3 Treeing All wrinkles shall be removed from the boots while on the last, and no material shall be used that might damage the leather or thread.

3.7.25.4 Final finish The boots shall be repaired and properly renovated. The materials and methods used shall be as specified in 3.3.16 through 3.3.18 inclusive. All exposed cut edges of the upper leather shall be stained to match the color of the leather.

3.7.26 Heelpads The heelpads specified in 3.3.4 shall be inserted grain side or buffed side up. The entire surface of the flesh side of the heelpads shall be coated with a suitable adhesive and firmly pressed into the heel seat area. The heelpads shall match the contour of the insole heel seat area and the width shall be sufficient to cover the entire heel seat area. The heel pads shall be graded properly for all sizes.

3.7.27 Mating and lacing The boots shall be properly mated for size and lefts and rights. A lace shall be inserted through the top peed lacer on the inside quarter of each boot. Adjust the lace into two equal lengths, slide on the clinch clamp to a point approximately one inch from the speed lacer and lock the clamp. Join the two boots together by tying the laces in a bow loop.

3.8 Marking and labeling

3.8.1 Contract marking The contractor's name, number and date of contract, and the proper size and width of each boot shall be lightly stamped with a permanent marking ink of contrasting color on either quarter lining. The markings shall be centered approximately 3/4 inch from the top. The maximum height of numbers and letters shall be 1/4 inch.

3.8.2 ANSI labeling The boots shall be labeled to indicate that they conform to Class 75 safety-toe requirements outlined in American National Standards Institute Standard ANSI Z41.1. Each boot shall have an indelible cloth label stitched to the inside of the boot on one of the quarter linings. The numbers and letters shall be a minimum of 3/16 inch in height and the label inscription shall read as follows:

ANSI
Z41.1-1967/75

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3.8.3 Instruction tag for lacing and clinching The instruction tag shall be fabricated from commercial grade tag stock and shall measure 2 by 3 inches. One corner of the instruction tag shall be perforated to receive the lace. The following information shall be printed in suitable size lettering, in black ink, on both sides of the instruction tag.

LACING AND CLINCHING INSTRUCTIONS

1. Single knot lace at mid center prior to installing in boot.
2. Lace up boot; insert ends of lace through clinch clamp; knot ends of lace together approximately one inch from tips.
3. Don boot; grasp lace ends, and pull up to tighten lacing.
4. Slide clinch clamp down lace to boot top and lock.
5. Pull up cuff; wrap lace around top of boot, storing lace between top of hook fastener tape and top of boot. Turn cuff down, and press cuff firmly to fasten hook and pile fastener tape.

3.9 Leakage The finished boots shall show no leakage when tested as specified in 4.3.5.2.

3.10 Workmanship The finished boots shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection The first article submitted in accordance with 3.2 shall be inspected as specified in 4.3.5 for compliance with design, construction, workmanship and dimensional requirements.

4.3 Quality conformance inspection Sampling for inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

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4.3.1 Component and material inspection In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, testing shall be performed on components and materials listed in Table VI. When data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in Table VI they shall be reported as required by the referenced test methods. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units or the composite sample fail to meet any requirements specified. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 thru 22,000	3
22,001 and over	5

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TABLE VI - TESTING OF COMPONENTS

Component and unit of product	Characteristic	Specification reference		Requirement applicable to		No. of determinations per sample unit	Results reported as	Sample unit
		Requirement paragraph	Test method	Sample unit	Lot average			
Upper leather (square feet)	Tannage type, thickness, grain finish	3.3.1	1/	-	-	-	-	One piece 8x8 2/
	Water penetration	3.3.1.1.1	ASTM D-2098 D-2099	X	-	1	No. of cycles	15 specimens"a" area 2/ 8 specimens"x" area
	Water absorption	3.3.1.1.2	4.4.1	X	-	3	nearest %	15 specimens"a" area 8 specimens"x" area 2/
	paranitrophenol content (when applicable)	3.3.1.2	FED-STD-311 6711 1/	-	X	-	-	
Lining leather (square feet)	Type, tannage, grain, thickness	3.3.3	1/					
Heel pads	Type, thickness and finish	3.3.4	1/					
Insoles, synthetic (pairs)	Mat'l Ident Copper 8-quin Thickness	3.3.5.2 3.3.5.2 3.3.5.2	1/ 1/ 1/					
Vinyl welting (yds)	Mat's Ident. Hardness (initial) Tensile strength (initial	3.3.6 3.3.6 3.3.6	1/ ASTM D-2240 3/ D-412 (DIE-E)	- x X	- - -	- 3 5	- number nearest 10 psi	5 yards

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TABLE VI-TESTING OF COMPONENTS

Component and unit of product	Characteristic	Specification reference		Requirement applicable to		No. of determinations per sample unit	Results reported as	Sample unit
		Requirement paragraph	Test method	Sample unit	Lot average			
Vinyl welting (yds) (cont'd)	Elongation (initial)	3.3.6	ASTM D-412 (DIE-E)	X	-	5	nearest 5%	
	Specific gravity	3.3.6	4.4.2.1	X	-	3	nearest .01%	
	Water absorption	3.3.6	4.4.2.2	X	-	3	nearest .01%	
	Brittle point, before and after ageing	3.3.6	4.4.2.3 and D-746	X	-	5	nearest 1°F	
	Hardness (after immersion)	3.3.6	4.4.2.4.1 and D-2240 3/	X	-	3	number	
	Tensile strength (after immersion)	3.3.6	4.4.2.4.2 and D-412 (DIE-E)	X	-	5	nearest 10 psi	
	Elongation (after immersion)	3.3.6	4.4.2.4.2 and D-412 (DIE-E)	X	-	5	nearest 5%	
	Volume swell	3.3.6	4.4.2.4.3	X	-	3	nearest 1%	
Counters (alternate)	Thickness	3.3.7	1/	-	-	-	-	
	Specific gravity	3.3.7	ASTM D-792	X	-	3	nearest .01%	
	Tensile streng.	3.3.7	ASTM D-1708	X	-	5	nearest 10 psi	
	Elongation	3.3.7	ASTM D-1708	X	-	5	nearest 5%	
	Flexural modulus	3.3.7	ASTM D-790	X	-	5	nearest 50 psi	
	Brittleness	3.3.7	ASTM D-746	X	-	5	nearest 2°	

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TABLE VI - TESTING OF COMPONENTS

Component and unit of product	Characteristic	Specification reference		Requirement applicable to		No. of determinations per sample unit	Results reported as	Sample unit
		Requirement paragraph	Test method	Sample unit	Lot average			
Doublers, cotton flannel (yds)	Mat'l Ident.	3.3.9.1	1/	-	-	-	-	
	Weight	3.3.9.1	FED-STD-191 5041	X	-	5	-	-
	Thread count	3.3.9.1	5050	X	-	5	-	One yard full width
	Breaking strength	3.3.9.1	5100	X	-	5	-	
	Non-fibrous mat'l	3.3.9.1	2611	X	-	1	-	-
	Color	3.3.9.1	1/	-	-	-	-	-
Alternate Toe cushion (feet)								
Polyvinyl Foam	Mat'l Ident.	3.3.9.2 (a)	1/					
	Thickness	3.3.9.2	1/					
	Weight	3.3.9.2	1/					
Latex foam	Mat'l Ident.	3.3.9.2 (b)	1/					
	Thickness	3.3.9.2	1/					
Understructure Box toe (feet)	Mat'l Ident.	3.3.9.3.2	1/					
	Thickness	3.3.9.3.2	1/					
Steel box toe (pair)	Mat'l Ident.	3.3.10.1	1/	-	-	-	-	2 box toes
	Hardness	3.3.10.1	FED-STD-151 Method 243	X	-	3	number	-
	Carbon content	3.3.10.1	1/	-	-	-	-	-
	Coating Ident.	3.3.10.1	1/	-	-	-	-	-
	Thickness	3.3.10.1	1/	-	-	-	-	-

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TABLE VI - TESTING OF COMPONENTS

Component and unit of product	Characteristic	Specification reference		Requirement applicable to		No. of determinations per sample unit	Results reported as	Sample unit
		Requirement paragraph	Test method	Sample unit	Lot average			
Eyelets (pound)	Mat'l Ident.	3.3.10.2	1/	-	-	-	-	2 eyelets
	Thickness	3.3.10.2	1/	-	-	-	-	-
	Finish	3.3.10.2	1/	-	-	-	-	-
	Diameter	3.3.10.2	1/	-	-	-	-	-
Speed lacers	Mat'l Ident.	3.3.10.3	1/					
	Finish	3.3.10.3	1/					
Cinch clamp	Mat'l Ident.	3.3.10.4	1/					
Heel nails (pounds)	Mat'l Ident.	3.3.10.5	1/	-	-	-	-	-
Tacks and staples (pounds)	Mat'l Ident.	3.3.10.6	1/	-	-	-	-	-
Nails, heel Seat Fastening (pounds)	Type of head	3.3.10.7	1/	-	-	-	-	-
	Mat'l Ident.	3.3.10.7	1/	-	-	-	-	-
Steel shank (shanks)	Mat'l Ident.	3.3.10.8.1	1/ FED-STD-151	-	-	-	-	-
	Hardness	3.3.10.8.1	Method 243	X	-	3	number	3 shanks
	Thickness	3.3.10.8.1	1/	-	-	-	-	-
	Coating Ident.	3.3.10.8.1	1/	-	-	-	-	-
	Coating thickness	3.3.10.8.1	1/	-	-	-	-	-
	Width	3.3.10.8.1	1/	-	-	-	-	-
Shankboard (boards)	Mat'l Ident.	3.3.10.8.2	1/	-	-	-	-	-
	Thickness	3.3.10.8.2	1/	-	-	-	-	-

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TABLE VI - TESTING OF COMPONENTS

Component and unit of product	Characteristic	Specification reference		Requirement applicable to		No. of determinations per sample unit	Results reported as	Sample unit
		Requirement paragraph	Test method	Sample unit	Lot average			
Fiberglass shank	Mat'l Ident.	3.3.10.9	1/	-	-	-	-	-
	Width	3.3.10.9	1/	-	-	-	-	-
	Heat curable	3.3.10.9	1/	-	-	-	-	-
Bottom filler- Thermoplastic	Mat'l Ident.	3.3.12.1	1/	-	-	-	-	-
	Mixture	3.3.12.1	1/	-	-	-	-	-
	Solubility	3.3.12.1	1/	-	-	-	-	-
	Softening point	3.3.12.1	1/	-	-	-	-	-
	Penetration	3.3.12.1	1/	-	-	-	-	-
Bottom filler Cold Process	Mixture	3.3.12.2	1/	-	-	-	-	-
	Solubility	3.3.12.2	1/	-	-	-	-	-
	Softening point	3.3.12.2	1/	-	-	-	-	-
Seam seal adhesive, vinyl (gallon)	Mat'l Ident.	3.3.14.1.1	1/	-	-	-	-	-
	Gallon weight	3.3.14.1.1	ASTM D-816 1/	X	1	-	ounce	one quart
	Total solids	3.3.14.1.1	4.4.6	X	2	-	0.1 percent	
	Viscosity	3.3.14.1.1	ASTM D-1084	X	2	0	1 second	
Adhesive general	Mat'l Ident.	3.3.14.2	1/	-	-	-	-	-
Stuck-on rib fabric	Mat'l Ident.	3.3.19	1/	-	-	-	-	-
	Weight	3.3.19	1/	-	-	-	-	-
	Yarns per inch (warp and filling)	3.3.19	1/	-	-	-	-	-
	Breaking streng.	3.3.19	1/	-	-	-	-	-

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.
2/ Sampling procedures and locations from which the sample unit is to be obtained shall be in accordance with ASTM STANDARD D-2813.
3/ The immersion medium shall be fluid III of TT-S-735.

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4.3.2 Examination of components

4.3.2.1 Examination of upper leather cut parts prior to fitting A 100% examination of each cut vamp, quarter, tongue, and back stay for defects listed below shall be conducted prior to fitting. Any cut vamp, quarter, tongue, or back stay containing one or more such defects shall result in a rejection of that cut part.

Scratch
Slaughter cut
Pronounced veins
Cuts on grain surface of leather
Brands or light flesh cuts that show through on the grain surface
Flanky, boney, loose, pipey, cracked, fleshy or coarse grained leather

4.3.2.2 Examination of insoles
(a) Leather insoles - When used, the leather insoles shall be examined for visual and dimensional requirements conforming to the provisions of the end item examination outlined in KK-I-570.

(b) Synthetic insoles When used, the synthetic insoles shall be examined for the defects listed below. The inspection level shall be level II and the acceptable quality level (AQL) shall be 4.0 defects per hundred units.

Any hole, cut, tear or gouge.
Any brittle area or evidence of delamination.
Thickness not as specified.
Sole outline not conforming to required pattern.
Sole not cut in specified direction.

4.3.3 In-Process inspection Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are being carried out as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection had indicated nonconformance.

4.3.4 Intermediate inspection

4.3.4.1 Visual examination The defects found during the intermediate examination shall be classified in accordance with 4.3.4.1.1 and 4.3.4.1.2. The applicable inspection levels and acceptable quality levels shall be as indicated in 4.3.4.1.3. NOTE: Defects designated by an asterisk (*) shall be scored as major when seriously affecting serviceability and minor when affecting serviceability, but not seriously.

4.3.4.1.1 Examination of uppers after all fittings The upper assembly shall be examined for defects in cutting, fitting and other construction characteristics which cannot be seen in the end item. The sample unit shall be one completely fabricated upper assembly prepared for lasting. The lot size shall be expressed in the same terms as the sample.

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Examine	Defect	Classification Major	(*)	Minor
Construction and workmanship (general)	a. Construction not as specified	X		
	b. Any component missing or other than type specified	X		
	c. Any component misplaced or not affixed as specified		*	
	d. Any component not skived as specified	X		
	e. Vamp and doubler not crimped			X
	f. Tongue not creased			X
	g. Backseam not rubbed down			X
	h. Backseam tape not caught in both rows of stitching			X
	i. Edges of vamp, quarter, backstay and gusset not seam sealed	X		
	j. Inside stitching of vamp throat and barring not coated or seam sealed	X		
	k. Outside upper threads not coated	X		
	l. Inside upper stitching not seam sealed (when specified)	X		
	m. ANSI label missing 1/			
Quality of leather (upper and lining)	a. Not full grain	X		
	b. Not specified color	X		
	c. Coarse, rough fiber on the flesh side		*	
	d. Slaughter cut or otherwise damaged	X		
	e. Thickness more than 1/2 ounce less than the minimum specified	X		
	f. Thickness more than the maximum or up to 1/2 ounce less than the minimum specified			X

1/ The contractor shall perform 100% examination for this defect. Any defective unit found shall be repaired, replaced or excluded from the in-process lot.

4.3.4.1.2 Examination of boot before bottom filling The partially fabricated boot shall be examined for defects in construction and workmanship. The sample unit shall be one partially constructed boot at a point after lasting and attachment of shank but before the application of the bottom filler. The lot size shall be expressed in terms of the sample unit.

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Examine		Defect	Classification Major (*)	Minor
Bottom of boot	a.	Any component missing or other than specified type	X	
	b.	Shank not properly positioned e.g., end of steel shank extending beyond ball line	*	
	c.	Shank wrong size or malformed	*	
	d.	Shank not fitting contour of boot bottom	*	
	e.	Fiberglass shank not cured	X	
	f.	Shank not securely attached		X
	g.	Any insole or anchor tack or staple not removed	X	
	h.	Upper damaged in lasting	*	
	i.	Excess leather in heel seat interfering with proper fit of shank		X
	j.	Poor heel seat lasting e.g., heel seat not flat, upper not completely caught in lasting tacks		X
	k.	Poor side lasting, upper loose on last i.e, insufficient lasting staples to hold upper to last	*	
	l.	Inseam not properly trimmed	*	
	m.	Inseam seal operation before and after welting omitted or not properly performed	X	
	n.	Less than 3 stitches per inch on inseam	X	
	o.	Less than 3 1/4 stitches but not less than 3 stitches per inch on inseam		X
	p.	Any inseam stitch broken, skipped cut or damaged	X	
	q.	Two or more inseam stitches not at bottom of insole rib or not in welt groove	*	
	r.	Broken insole rib	X	
	s.	Any operation omitted	X	
	t.	Any operation improperly performed	*	
	u.	Welt butt not properly skived and tacked		X
	v.	Ends of counter not caught by inseam stitching		X
Upper part of boot	a.	Uppers not firmly down to last		X
	b.	Lace opening more or less than specified		X
	c.	Quarters not laced in three or four lower eyelets		X
Steel toe sembly		Steel toe out of alignment or crooked	*	

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4.3.4.1.3 Acceptable quality levels (AQL's) and inspection levels The quality levels, expressed in defects per 100 units and the inspection levels shall be as follows:

	Inspection level	AQL's Major	Total
For intermediate examination (4.3.4.1.1 - fittings)	II	2.5	6.5
For intermediate examination (4.3.4.1.2 - before bottom filler)	II	2.5	6.5

4.3.4.2 Testing of combined rib and insole The stuck-on rib shall be tested for compliance with the requirements of 3.3.19. Method 2061 of FED-STD-311 shall be used for the shear strength test. Method 2171 of FED-STD-311 shall be used for the stitch tear strength. The sample unit shall be two ribbed insoles, one for each test, and the sample size shall be 5, regardless of the lot quantity. Requirements are applicable to the sample unit for minimum unit and lot average requirements. The lot shall be unacceptable if one or more sample units or the lot average fail to meet the specified requirement. All test reports shall contain the individual values utilized in expressing the final result.

4.3.5 Examination of the end item The defects found during the examination of the end item shall be classified in accordance with 4.3.5.1. The applicable inspection level and acceptable quality levels (AQL's) shall be as indicated in 4.3.5.1.1. In addition, the finished boots shall be tested for leakage in accordance with 4.3.5.2 and 4.4.9.

4.3.5.1 Visual examination The boots shall be examined for defects listed below. The sample unit shall be one completely fabricated boot except that laces need not be inserted and the selection shall be by pairs. Heel pads removed during Government verification inspection shall be re-cemented and replaced by the contractor. For pairing examination, the pair shall be examined together. Each defect found during the examination for pairing shall be scored as a single defect. The lot size shall be expressed in units of one boot. The vamp and quarters shall be examined for break of leather in accordance with the procedure.

Vamp Any vamp exhibiting a break pattern that is greater than that represented by number 5 on the Satra Scale (see 6.4) shall be scored a defect regardless of the direction in which the break pattern appears. To examine vamp the boot shall be held in an upright position with both hands. The toe of the boot shall face away from the examiner. Position thumbs on top of vamp approximately half way between the box toe line and blucher noses and 1 to 1 1/2 inches apart. Press downward with thumbs and observe break pattern across the surface of leather.

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Quarters: Bend each quarter in a concave position in any direction. The break in leather shall be compared with the break pattern of the Satra Scale (see 6.4). Any quarter exhibiting a break pattern greater than No. 5 on the Satra Scale shall be scored as a defect.

NOTE: Asterisk defects listed below shall be scored as "Major" when affecting serviceability or appearance seriously and "Minor" when affecting serviceability or appearance but not seriously.

Examine	Defect	Classification Major	(*)	Minor
Pairing	a. Not properly mated, i.e., not right and left of same size	X		
	b. Variation in color or appearance		*	
Cleanliness	Any spot, stain or foreign matter			X
Color or finish	a. Not specified color	X		
	b. Color not uniform			X
	c. Any raw edges not stained to match upper leather			X
	d. Finish flaky, chipped, streaky or shows runs on uppers			X
Design, type and size	Not as specified	X		
Material (general)	Any component not fabricated of the material specified	X		
Upper leather	a. Not full grain	X		
	b. Fat wrinkles, pits, insect damage	X		
	c. Pipey, boney, loose, flanky, stretchy	X		
	d. Leather damaged in processing		*	
	e. Brands, open scars, fleshy, slaughter cuts, cracked, coarse deep scratches		*	
	f. Break of vamp or quarters greater than No. 5 on SATRA SCALE	X		
	NOTE: Leather showing a break pattern greater than No. 5 is considered to be loose, pipey, or flanky.			
Lining leather	a. Wrinkled or excessive fullness of lining		*	
	b. Flesh cuts	X		
	c. Flanky or pipey	X		

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Examine	Defect	Classification Major	(*)	Minor
Construction and workmanship (general)	a. Any cut, tear, hole, repair or abrasion		*	
	b. Any component or assembly misplaced, operation omitted or not properly performed, e.g., backstay or blucher noses crooked to a degree where it is readily noticeable		*	
	c. Wrinkled or bunched area at backseam		*	
	d. Any component or assembly misplaced or operation omitted or not properly performed		*	
Seams and stitching (upper)	a. Any open seam (A seam shall be classified as open when one or more stitches joining a seam are broken, or when two or more consecutive stitches or runoffs occur. On multiple stitched seams, a seam is considered open when one or more rows of stitching are open)		*	
	b. Loose tension resulting in loosely secured seam		*	
	c. Tight tension resulting in puckering or cutting of leather		*	
	d. Wrong stitch type or seam type	X		
	e. Less than the specified minimum number of stitches		*	
	f. More than the specified maximum number of stitches:			
	(1) resulting in damage to leather	X		
	(2) but does not damage leather			X
	g. Gage of stitching not as specified or irregular			X
	h. Any row of stitching omitted		*	
	i. Thread ends not trimmed throughout boot			X
	j. Needle holes or needle chews			X

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Examine	Defect	Classification Major	(*)	Minor
Outsole stitching and bottom attaching (Goodyear)	a. Lock not just under surface of outsole	X		
	b. Lock on surface of outsole	X		
	c. Less than 3 1/2 stitches per inch	X		
	d. Less than 4 1/2 but not less than 3 1/2 stitches per inch			X
	e. More than 7 stitches per inch, except ball and toe sole stitching			X
	NOTE: More than 3 1/2 stitches per any one half length in the ball and toe sole stitching shall be classified as a minor defect. The ball and toe sole stitching shall be defined as that portion of the sole stitching that runs from the inside ball and around the toe to the outside ball.			
	f. Goodyear stitching running into tread design	X		
	g. Stitches do not extend under heel breast heel		*	
	h. Skipped or broken stitch		*	
	i. Goodyear stitching cutting into sole substance or deeper than specified	X		
	j. Gauge of Goodyear stitching not as specified or irregular			X
	k. Checked sole, i.e., separation of sole and welt		*	
	l. Any stitching not visible on surface of sole	X		
Counters	a. Counters not filling counter pocket area		*	
	b. Rolled or curled counter		*	
Edge making	a. Edge trimmed into Goodyear stitching	X		
	b. Edge not trimmed square or trimming is irregular			X

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Examine	Defect	Classification Major (*)	Minor
Edge making (cont'd)	c. Sole extension less than the specified minimum:		
	(1) By not more than 3/32 inch		X
	(2) By more than 3/32 inch	X	
	d. Sole extension is beyond specified maximum		X
Heel finishing and attaching	a. Any visible floating washer	*	
	b. Heel not finished square, i.e. flared or tapered more than 1/8 inch		X
	c. Heel not finished smooth		X
	d. Checked heel, i.e., separation of heel and outsole		X
	e. Wrong size or type heel		X
	f. Breast of heels not beveled		X
	g. Any heel nail missing		X
	h. Open-heel seat	*	
	i. Heel crooked		X
	j. Either corner breast nail not properly clinched on insole	X	
	k. Two or three nails other than corner breast nail, not properly clinched on insole		X
	l. More than three nails other than corner breast nails, not properly clinched on insole	X	
	NOTE: Evidence of nails on insole shall not be interpreted as proper or sufficient clinching.		
Lacers and eyelets	a. Number of lacers or eyelets not as specified	X	
	b. Lacers not evenly spaced or aligned within each row		X
	c. Misalignment between the rows to the extent it interfers with proper lacing	X	
	d. Edge of bottom eyelet less than 3/16 inch or more than 5/16 inch from edge of quarter		X
	e. Lacer or eyelet not securely clinched		X
	f. Lacer deformed, does not function properly	X	
	g. Outer edge of lacer ring does not finish even with or projects more than 1/8 inch beyond edge of quarters		X
	h. Top lacer or bottom eyelet not positioned as specified		X

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Examine	Defect	Classification Major	(*)	Minor
Inseaming	a. Grinning seam, i.e., thread exposed	X		
	b. Strained seam, i.e., needle holes visible but thread not exposed			X
Insole	a. Short or long		*	
	b. Any protruding point of tack, staple, or nail on insole in heel area	X		
	c. Any protruding point of nail, staple, or tack on insole forward of the heel breastline (see footnote 1/)			
Steel box toe	Missing (see footnote 2/)			
Marking and labeling	a. Missing, incomplete, incorrect, not applied in the specified manner, misplaced, illegible, or not specified size			X
	b. Missing ANSI label 3/			

1/ Any protruding point of tack, staple, or nail found in the sample shall cause rejection of the lot presented.
2/ Any missing steel box toe found in the sample shall cause rejection of the lot presented.
3/ Any missing ANSI label found in the sample shall be scored and the boot repaired, replaced or excluded from the lot.

4.3.5.1.1 Acceptable quality levels (AQL's) and inspection level The acceptable quality levels, expressed in defects per 100 units, and the inspection levels shall be as follows:

	Inspection level	AQL's Major	Total
For end item examination (4.3.5.1)	II	2.5	6.5

4.3.5.2 Testing of the end item . The end item shall be tested for the performance characteristics listed in Table VII. The sample unit for all tests shall be one boot. The lot size shall be expressed in terms of one boot. The inspection level shall be S-2. The AQL for the water resistant test shall be 6.5 defects per hundred units (DHU). Failure of any sample to pass the impact requirement shall result in rejection of the entire lot represented by the sample. Requirements for all tests are applicable to the sample unit.

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Table VII - End item testing

Sample Unit	Characteristic	Requirement Paragraph	Test Method	Determination per sample unit	Results: reported as
One boot	Impact	3.3.10.1.1	4.4.4	1	Pass or Fail
One boot	Leakage	3.9	4.4.5	1	Pass or Fail

4.3.6 Packaging inspection An examination shall be made to determine that preservation-packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully packaged. The lot size shall be the number of containers in the end item inspection lot. The inspection shall be S-2 and the AQL shall be 2.5 defects per one hundred units.

Examine	Defect
Marking (exterior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application. Size marking on boots does not conform to the size shown on exterior container. 1/
Materials	Any component missing, damaged or not as specified.
Workmanship	Inadequate application of components such as: incomplete closure of container flaps, loose strapping, inadequate stapling. Bulged or distorted container.
Content	Number of pairs of boots per container is more or less than specified.

1/ For this defect, one pair of boots from each container in the inspection lot shall be examined.

4.4 Tests.

4.4.1 Test for water absorption of upper leather. The upper leather shall be sampled and conditioned in accordance with ASTM standards D-2813 and D-1610. Three 4 by 4 inch test specimens shall be completely immersed in 1000 ml beaker of distilled water 23° (± 1°C) in such a manner that the surfaces of the specimens do not touch each other. After one hour (± 1) minute immersion, the specimens shall be withdrawn from the beaker and the surface water of each removed by lightly blotting the specimens with blotting paper prior to weighing. Each specimen shall then be weighed immediately and the weight recorded as W_2. The amount of water absorbed by each specimen shall be calculated as follows:

$$\text{Water absorbed, percent} = \frac{W_2 - W_1}{W_1} \times 100$$

Where: W_1 = weight of the original specimen, grams and W_2 = weight of specimen after immersion in water, grams.

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4.4.2 Test for vinyl welting The testing of the vinyl welting shall be performed as specified in table VI and 4.4.2.1 thru 4.4.2.6.

4.4.2.1 Specific gravity The specific gravity of the vinyl welting shall be determined by a Fisher Young Gravitometer using a specimen from 2 to 3 inches in length.

4.4.2.2 Water absorption A specimen of vinyl welting shall be cut to a length of 4 (± .05) inches, weighed to the nearest .001 gram (W_1) immersed in a beaker of distilled water and maintained for seven days in an oven at a temperature of 70° (± 2°)C. The beaker shall then be removed from the oven and cooled to room temperature. The test specimen shall then be washed in methanol, dried for one minute and reweighed (W_2). The top of the beaker shall be sealed during the test to prevent evaporation of the distilled water. The percentage of water absorbed shall be calculated from the following formula:

$$\text{Water absorption percent} = \frac{W_2 - W_1}{W_1} \times 100$$

4.4.2.3 Brittle point after ageing A sample of vinyl welting shall be placed in an oven and maintained at a temperature of 100° (± 2°)C for seven days. The sample shall then be removed and allowed to remain at room temperature for 24 hours prior to testing as specified in Table VI.

4.4.2.4 Solvent resistance.

4.4.2.4.1 Hardness reading. A specimen of Vinyl welting, a minimum length of 3 inches of the full width and thickness of the welting, shall be immersed at a temperature of 23 ± 2 degrees Centigrade for 24 ± 1/4 hours in type III fluid of TT-S-735. The specimen shall be completely immersed in the fluid and the container shall be fitted with an air or reflux condenser so as to maintain a constant volume. At the end of the immersion period, the specimen shall be removed from the fluid, blotted dry with filter paper or other absorbent material, and tested immediately for hardness as specified in Table VI.

4.4.2.4.2 Elongation and tensile strength A specimen of the vinyl welting, 6 by 0.45 by 0.125 (± 3 percent) inches shall be completely immersed at a temperature of 23 (± 2) degrees Centigrade for 24 (± 1/4) hours in type III fluid of TT-S-735. The container shall be fitted with an air or reflux condenser so as to maintain a constant volume. At the end of the immersion period, the specimen shall be removed from the fluid, blotted dry with filter paper or other absorbent material and hung up in a temperature of 23 (± 2) degrees Centigrade for 24 (± 1/4) hours and then tested for elongation and tensile strength as specified in Table VI.

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4.4.2.4.3 Volume swell A 50 ml. burette calibrated 0.1 ml shall be filled with 30 ml. (V1) of methanol. The test specimen shall be 1/4 (± 1/16) inch in length which shall be immersed in the methanol, and the increase volume shall be recorded as (V2). The specimen shall then be removed, dried and inserted in the burette filled with 30 ml. (V1b) of methanol, and the increased volume shall be recorded as (V2b). The percent of volume swell shall be calculated from the following formula:

$$\text{Volume swell percent} = \frac{\begin{array}{l} V2-V1 = VA \\ V2b-V1b = VB \\ VB-VA \end{array}}{VA} \times 100$$

4.4.3 Gaging of nail points The gaging of nail points shall be done with an English standard wire gauge in which 20 gage is 0.035 (± 0.005) inch. The points of the nails shall enter the gage at least 1/32 inch.

4.4.4 Test for impact resistance of steel box toe The impact test shall be performed in accordance with the following procedure: A falling weight of steel or other suitable material, weighing not less than 49 1/2 pounds nor more than 50 1/2 pounds, 1 inch (± 0.020 inch) in diameter and with a striking face made hemispherical in shape by rounding on a 1 inch radius, equipped to drop freely in a tube or other suitable guide, shall be used. The weight shall be dropped from a distance of 18 inches, (± 1/4 inch), above the top of the box toe. As an alternate, a falling weight of 25 (± 1/4) pounds shall be dropped from a distance of 36 (± 1/2) inches. The equipment shall be assembled so that the center of striking face of the falling weight will strike the boot at midwidth 1/2 inch in front of the back edge of the box toe. The boot (or toe-section) shall be mounted on an anvil (or rigidly mounted bedplate) of steel or other suitable material weighing not less than 500 pounds, equipped with suitable guides (or clamps) to hold the boot specimen in place. In performing the test, the minimum inside clearance of 1/2 inch may be determined instantaneously by placing lumps of moldable material such as wax or modeling clay inside the boot directly below the point of impact. The moldable material should be compressed with the fingers so that it contacts both upper and lower inside surfaces of the boot before the test is made. When in place, the greatest horizontal dimension of the lumps should not exceed 1 inch. An approved method for impact testing is contained in American National Standard ANSI Z41.1-1976. The steel box toe shall be considered a failure if the moldable material is compressed to a point less than 1/2 inch in thickness.

4.4.5 Test for leakage resistance of boots A finished boot shall be immersed in water over the welting for a period of 15 minutes. No evident penetration of water on the insoles or lining leather including any evidence of water under the sealer coat shall be permitted during this period.

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4.4.6 Total solids of seam sealant

4.4.6.1 Apparatus

a. Flat bottom dishes, 70 mm in diameter of aluminum, such as for milk analysis or equal.
b. Analytical balance.
c. Laboratory oven, forced draft.

4.4.6.2 Procedure Place a portion of the thoroughly mixed sample in a dropping bottle and weigh to the nearest .001mg. Weigh aluminum dishes with fourth - decimal accuracy. Transfer a sample weight of two grams to a dish, determine the exact weight by loss in weight of the bottle. Place dish in the oven at 105°C for two hours. Upon cooling in a desicator, reweigh the dish to the nearest .001mg. From the weight of the residue in the dish and the weight of the sample taken, calculate the percent of total solids.

5. PACKAGING

5.1 Preservation-packaging Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A Each pair of properly mated boots shall be tied together as specified in 3.7.2.7.

5.1.2 Level C (Commercial packaging) Boots shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The package and quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A Six pairs of boots of one size and width only, preserved as specified in 5.1, shall be packed in a snug fitting fiberboard shipping container conforming to type CF, weather-resistant, style FOL-L, variety DW, grade V15c, of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather resistant, variety DW, grade V15c of PPP-B-636. The approximate inside dimensions of the shipping container are 19 3/4 inches in length, 14 3/4 inches in width, and 14 1/2 inches in depth. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Each container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced in accordance with the appendix of PPP-B-636. Toward the end of the contract or when there are less than the required amount per container of the same size and width, mixed sizes and widths may be packed in the same container.

5.2.2 Level B Six pairs of boots of one size and width only, preserved as specified in 5.1, shall be packed in a snug fitting corrugated fiberboard shipping container conforming to FOL-L, type CF, class domestic, grade 275, variety DW of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275

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of PPP-B-636. The approximate inside dimensions of the shipping container are 19 3/4 inches in length, 14 3/4 inches in width, and 14 1/2 inches in depth. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Each container shall be closed in accordance with Method II as specified in the appendix of PPP-B-636. Toward the end of the contract or when there are less than the required amount per container of the same size and width, mixed sizes may be packed in the same container.

5.2.3 Level C (Commercial packing) The boots packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Labels, mixed sizes Each shipping container packed with mixed sizes and widths, shall have securely attached to the side and end, a white paper label 5 by 4 inches, with the words MIXED SIZES AND WIDTHS clearly printed thereon and under these words shall be legibly printed the correct quantity of pairs, sizes and widths contained therein.

6. NOTES

6.1 Intended use The boots covered by this specification are intended to be worn for operational flight purposes by personnel of the Department of Defense.

6.2 Ordering data Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Sizes and widths required (see 1.2).
c. Whether first article sample is required (see 3.2).
d. Availability of lasts and patterns (see 3.5 and 3.6).
e. Selection of applicable levels of preservation-packaging and packing required (see 5.1 and 5.2).

6.3 Samples For access to samples, address the procuring activity issuing the invitation for bids.

6.4 Satra scale The Satra Scale may be obtained from the British Shoe and Allied Trade Research Association, Satra House, Kettering, Northants, England, or may be obtained from Bata Engineering, Batawa, Ontario, Canada.

6.5 Suppliers of component parts To obtain a list of approved component part suppliers write to: Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760. The list is available only to indicate possible sources of supply. Any unlisted supplier with a similar item that is equal to or better than the approved supplier's item will also be acceptable.

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6.6 Fiberglass shank A fiberglass shank manufactured by American Shoe Machinery Co., 30 Nashua Street, Woburn, MA 01801 under their designation "TRU-FIT" has been found to meet the requirements of 3.3.10.9.

Custodian:
NAVY - NU

Review Activity:
Navy - AS
DLA - CT

Preparing Activity:
Navy - NU

Project No. 8430-0328

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FIGURE 1 - BOOT (CUFF UP)

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FIGURE 2 - BOOT (CUFF DOWN OVER LACES)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL	MIL-B-24911

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-B-24911 Boots, Safety, Aircrew

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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